Lennard–Jones Potential and Reduced Units

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1 Lennard–Jones potential

Lennard-Jones potential is one of the most famous pair potentials for van der Waals systems, which was initially proposed for liquid argon (Ar).

$$V_{LJ} = 4\varepsilon \left[\left(\frac{\sigma}{r}\right)^{12} - \left(\frac{\sigma}{r}\right)^{6} \right] \tag{1}$$

The parameter ε governs the strength of the interaction and σ defines a length scale. The r^{-12} term, which is the repulsive term, describes Pauli repulsion at short ranges due to overlapping electron orbitals, and the r^{-6} term, which is the attractive long-range term, describes attraction at long ranges (van der Waals force). This potential is strongly repulsive at short distances and it passes through 0 at $r = \sigma$, $V_{LJ} = 0$.

Whereas the functional form of the attractive term has a clear physical justification, the repulsive term has no theoretical justification. It is used because it approximates the Pauli repulsion well and is more convenient due to the relative computing efficiency of calculating r^{-12} as the square of r^{-6}.

Due to its simplicity, it is often used to describe the properties of gases and to model dispersion and overlap interactions in molecular models. It is especially accurate for noble gas atoms and is a good approximation at long and short distances for neutral atoms and molecules.

These parameters can be fitted to reproduce experimental data or accurate quantum chemistry calculations. Due to its computational simplicity, the Lennard-Jones potential

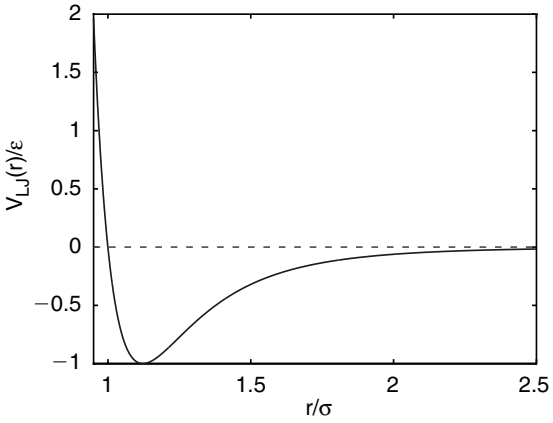


Figure 1: The Lennard–Jones potential.

Table 1. The L-J parameters between different molecules.

Molecules	σ (Å)	ε (meV)
Liquid-liquid (LL)	3.41	10.30
Liquid-gas (LG)	3.41	6.87
Liquid-solid at top (LS_{top})	3.41	5.15
Liquid-solid at bottom (LS_{bot})	3.41	5.66
Gas-gas (GG)	3.41	3.43
Gas-solid at top (LS_{top})	3.41	1.72
Gas-solid at bottom (LS_{bot})	3.41	1.89
Solid-solid (SS)	0	0

Table2: Units used in MD simulations of Lennard-Jones potential of Ar.

Physical quantity	Unit	Value for Ar
length	σ	3.4×10^{-10} m
energy	ε	1.65×10^{-21} J
mass	m	6.69×10^{-26} kg
time	$\sigma(m/\varepsilon)^{1/2}$	2.17×10^{-12} s
velocity	$(\varepsilon/m)^{1/2}$	1.57×10^{2} m/s
force	ε/σ	4.85×10^{-12} N
pressure	ε/σ^3	4.20×10^{7} N $\cdot$ m^{-2}
temperature	ε/k_B	120 K

2 Reduced Units

It is convenient to work in dimensionless or reduced units, also called MD units. In MD units $k_B = 1$ is considered.

Assume that the potential energy of the system can be written as a sum of pair interactions. The (effective) intermolecular potential can be written in the form $\varphi(r) = \varepsilon V(r/\sigma)$, where ε is a characteristic energy (*e.g.* the well depth) and σ a characteristic length (*e.g.* the value of r for which $\varphi(r) = 0$). The mass of the atoms is denoted by m. It is good practice to express all physical quantities in term of the units ε, σ and m.

In this way one avoids discovering the law of corresponding states the hard way. From these fundamental units one may derive all other units, *e.g.* the unit of time: $t_0 = \sigma(m/\varepsilon)^{1/2}$. To give a specific example: for Lennard-Jones (Ar) ($\varepsilon/k = 119.8$ K, $\sigma = 3.405$ Å, $m = 39.95/6.02 \cdot 10^{23}$ g) the time unit is $t_0 = 2.156 \cdot 10^{-12}$ s, the unit of velocity $v_0 = \sigma/t_0 = 1.579 \cdot 10^{4}$ cm $\cdot$ s^{-1}. The time unit t_0 is roughly the timescale of one atomic vibration period in Ar lattice.

Y. Liu and X. Zhang, J. Chem. Phys.**141**, 134702 (2014)

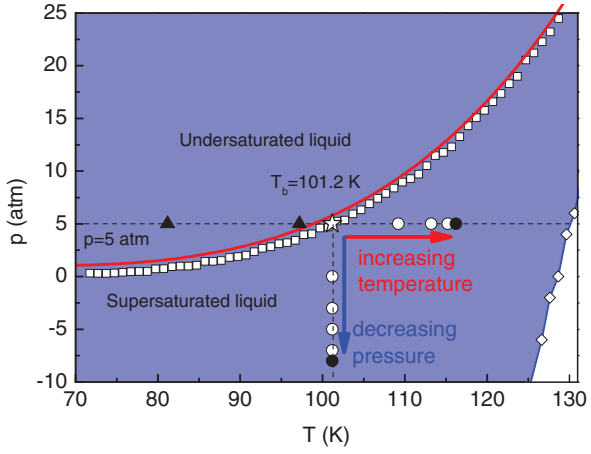


FIG. 2. The phase diagram for the pure fluid in the pressure-temperature plane. The boiling point is 101.2 K at 5 atm. The empty square symbols indicate the saturated pressure at different temperature from the simulation. The diamond symbols indicate the stability limit of the metastable liquid. Other symbols represent the states simulated to investigate the nanobubble stability in the pure fluid. The triangle symbols indicate the Wenzel state, the star symbol indicates the Cassie state, the empty circle symbols indicate the presence of stable nanobubbles, and the solid circle symbols indicate the occurrence of liquid-to-vapor phase transition. See Fig. 3 for typical snapshots.

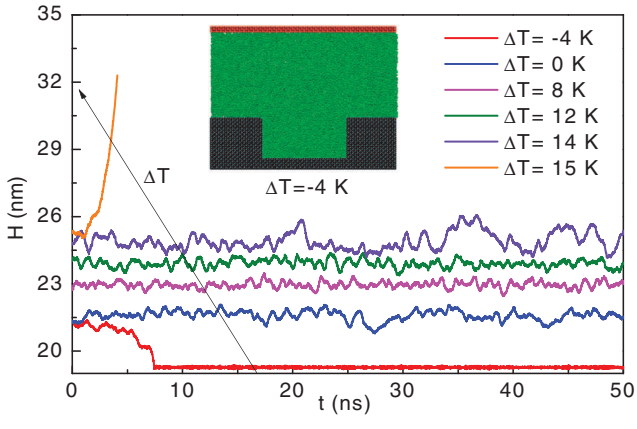


FIG. 3.Time evolution of the simulation box height at different temperatures. The inset figure shows the snapshot of final configuration at $\Delta T = -4$ K that corresponds to a Wenzel state.

P = 0.012173ε/σ^3 = 500 kPa = 5 atm; T = 0.846ε/k_B = 101.3 K; 1K = 0.00836ε/k_B;